2. 先证 $H \leq G$

① $H \neq \phi$，且 $H \subseteq G$

② $\forall h_1, h_2 \in H$. $h_1h_2 = \begin{pmatrix} 1 & t_1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & t_2 \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & t_1+t_2 \\ 0 & 1 \end{pmatrix} \in H$. 封闭

③ $h = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$，逆矩阵 $h^{-1} = \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix} \in H$.

再证 $H \lhd G$.

$\forall a = \begin{pmatrix} r & s \\ 0 & 1 \end{pmatrix} \in G$，$\forall h = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \in H$.

$a^{-1} = \begin{pmatrix} \frac{1}{r} & -\frac{s}{r} \\ 0 & 1 \end{pmatrix} \in G$.

有 $aha^{-1} = \begin{pmatrix} r & s \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{r} & -\frac{s}{r} \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & tr \\ 0 & 1 \end{pmatrix} \in H$

$\therefore H \lhd G$

3. 先证 $C(G) \leq G$

① $C(G) \neq \phi$ 且 $C(G) \subseteq G$

② $\forall a, b \in C(G)$. $\forall x \in G$,

$(ab)x = a(bx) = a(xb) = (ax)b = xab = x(ab)$

$\therefore ab \in C(G)$.

③ $\forall a \in C(G)$，$ax = xa$，

$\therefore a^{-1}(ax)a^{-1} = a^{-1}(xa)a^{-1} \Rightarrow xa^{-1} = a^{-1}x$

$\therefore a^{-1} \in C(G)$

再证 $C(G) \lhd G$

$\forall a \in G$，$\forall h \in C(G)$.

$aha^{-1} = aa^{-1}h = eh = h \in C(G)$.

$\therefore C(G) \lhd G$

6. ① 先证 $H \cap K \leq G$.

(i) $H \lhd G$，$K \lhd G$，$\therefore e \in H$，$e \in K$，

$\therefore e \in H \cap K$，即 $H \cap K$ 非空.

(ii) 封闭：$a, b \in H \cap K$，$ab^{-1} \in H \wedge ab^{-1} \in K$

$\therefore ab^{-1} \in H \cap K$.

(iii) 可逆. 同(ii)

再证 $H \cap K \lhd G$.

$\forall x \in H \cap K$，$\forall g \in G$，$x \in H$ 且 $H \lhd G$.

$gxg^{-1} \in H$. 同理，$gxg^{-1} \in K$，$\therefore gxg^{-1} \in H \cap K$.

$\therefore H \cap K \lhd G$

② $HK = \{h \cdot k \mid h \in H \wedge k \in K\}$

先证 $HK \leq G$. $\forall h_1k_1, h_2k_2 \in HK$，$\exists k_3, h_3$.

$(h_1k_1)(h_2k_2)^{-1} = (h_1k_1)(k_2^{-1}h_2^{-1}) = h_1(k_1k_2^{-1})h_2^{-1}$

$= h_1k_3h_2^{-1} = h_1(k_3h_2^{-1}) = h_1(h_3k_3) = (h_1h_3)k_3 \in HK$.

再证 $HK \lhd G$. $\forall g \in G$，$\forall hk \in HK$.

$ghg^{-1} \in H \wedge gkg^{-1} \in K$.

$ghkg^{-1} = gh(g^{-1}g)kg^{-1} = (ghg^{-1})(gkg^{-1}) \in HK$

$\therefore HK \lhd G$

8. 设 $G = \langle a \rangle$　$H \leq G$ 则 H 也为循环群.

设 $H = \langle a^m \rangle$，a^m 为 H 中 a 的最小正幂.

$\forall g \in G$，设 $g = a^i$，则有 $gH = Hg$，$\therefore H \lhd G$.

$\therefore \forall bH \in G/H$，$b = a^k$，$bH = a^kH = (aH)^k$

$\therefore G/H = \langle aH \rangle$